Consistency of restricted Cantor-Lukasiewicz naive set theory

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Editorial note. Official CC BY4.0 selected editable transcription; complete counted author construction and proof retained.

Editorial note. Source-specific grammar and wedge-right printed formulations remain unchanged; detailed source note discloses them.

Editorial note. Only source diagrams reformatted as editable fractions; no local compilation or mathematical certification.

Editorial scope. Complete author material collection for one structural theorem; author retains responsibility for the mathematics.

Allowed standard prerequisites (editorial).

- Brouwer fixed point theorem on a finite-dimensional compact hypercube. Original section4.3 explicitly applies BFT to its continuous finite formula-closure map.
- Standard Lukasiewicz algebraic semantics, finite strong completeness and continuity. Original Definitions1/2, Figure3 and Theorems10/12 retained; original standard literature[8,15,24,26] in complete numbered bibliography.

Author source: syntax and inference rules (30:3–4)

Let Var be a fixed countable set of propositional variables. The set Fm of formulas is defined by the following grammar:

$$A, B ::= p \mid 1 \mid 0 \mid A \wedge B \mid A \vee B \mid A \cdot B \mid A \to B \quad (p \in \mathsf{Var}).$$

Here we adopt the standard notation in substructural logics. In linear logic, the connectives $1, 0, \wedge, \vee, \cdot, \to$ are respectively written as $1, \bot, \&, \oplus, \otimes, \multimap$. We write $\neg A$ for $A \to 0$, and $A \leftrightarrow B$ for $(A \to B) \wedge (B \to A)$. A sequent is of the form $\Gamma \Rightarrow \Pi$, where Γ is a multiset of formulas and Π is either a singleton of a formula or an empty set. Inference rules of the sequent calculus for FLew are given in Figure 1.

$$\frac{}{A \Rightarrow A}\ (\text{init}) \qquad \frac{\Gamma \Rightarrow A \quad A, \Delta \Rightarrow \Pi}{\Gamma, \Delta \Rightarrow \Pi}\ (\text{cut})$$

$$\frac{\Delta \Rightarrow \Pi}{\Gamma, \Delta \Rightarrow \Pi}\ (i) \qquad \frac{\Gamma \Rightarrow}{\Gamma, \Delta \Rightarrow \Pi}\ (o)$$

$$\frac{A_i, \Delta \Rightarrow \Pi}{A_1 \wedge A_2, \Delta \Rightarrow \Pi}\ (\wedge\ell) \qquad \frac{\Gamma \Rightarrow A_1 \quad \Gamma \Rightarrow A_2}{\Gamma \Rightarrow A_1 \cdot A_2}\ (\wedge r)$$

$$\frac{\Delta \Rightarrow \Pi}{1, \Delta \Rightarrow \Pi}\ (1\ell) \qquad \frac{}{\Rightarrow 1}\ (1r)$$

$$\frac{A_1, \Delta \Rightarrow \Pi \quad A_2, \Delta \Rightarrow \Pi}{A_1 \vee A_2, \Delta \Rightarrow \Pi}\ (\vee\ell) \qquad \frac{\Gamma \Rightarrow A_i}{\Gamma \Rightarrow A_1 \vee A_2}\ (\vee r)$$

$$\frac{}{0 \Rightarrow}\ (0\ell) \qquad \frac{\Gamma \Rightarrow}{\Gamma \Rightarrow 0}\ (0r)$$

$$\frac{A, B, \Delta \Rightarrow \Pi}{A \cdot B, \Delta \Rightarrow \Pi}\ (\cdot\ell) \qquad \frac{\Gamma \Rightarrow A \quad \Delta \Rightarrow B}{\Gamma, \Delta \Rightarrow A \cdot B}\ (\cdot r)$$

$$\frac{\Gamma \Rightarrow A \quad B, \Delta \Rightarrow \Pi}{\Gamma, A \to B, \Delta \Rightarrow \Pi}\ (\to \ell) \qquad \frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \to B}\ (\to r)$$

Figure 1. Sequent calculus for FLew.

Given a formula $A \in \mathsf{Fm}$, we write A^{seq} for the sequent $\Rightarrow A$ with empty antecedent. Given a set $\Phi \cup \{A\}$ of formulas, we write $\Phi \vdash_{\mathsf{FLew}} A$ if A^{seq} is derivable from some sequents in $\Phi^{\mathrm{seq}} := \{B^{\mathrm{seq}} \mid B \in \Phi\}$ by rules in Figure 1.

Definition 1. A substructural logic over FLew is a set $L \subseteq \mathsf{Fm}$ which is closed under deduction in FLew and substitution:

$$L \vdash_{\mathsf{FLew}} A \Longrightarrow A \in L, \qquad A \in L \Longrightarrow A' \in L, \quad \text{where } A' \text{ is a substitution instance of } A.$$

We denote by $\mathrm{Ext}(\mathsf{FLew})$ the set of substructural logics over FLew.

Author source: algebraic semantics (30:4–5, 30:9)

Definition 2. An FLew algebra is a tuple $P = (P, \wedge, \vee, \cdot, \to, 1, 0)$ such that

- $(P, \wedge, \vee, 1, 0)$ is a bounded lattice with 1 greatest and 0 least,
- $(P, \cdot, 1)$ is a commutative monoid, and
- $a \cdot (-) : P \to P$ is a left adjoint of $a \to (-) : P \to P$ for every $a \in P$:

$$a \cdot x \leq y \Longleftrightarrow x \leq a \to y,$$

where $x \leq y$ is a shorthand for $x \vee y = y$ (or equivalently $x = x \wedge y$).

A formula $A \in \mathsf{Fm}$ can be seen as a term of FLew algebras. Hence it makes sense to consider a homomorhism $v : \mathsf{Fm} \to P$ for any FLew algebra P. Such v is called a valuation. We say that v satisfies $\Phi \subseteq \mathsf{Fm}$ if $v(A) = 1$ holds for every $A \in \Phi$. Given a class K of FLew algebras and a set $\Phi \cup \{A\}$ of formulas, we write $\Phi \models_K A$ if for every $P \in K$ and every valuation $v : \mathsf{Fm} \to P$ with $P \in K$, if v satisfies Φ, then it also satisfies A. When $\Phi = \varnothing$, we write $\models_K A$. When $K = \{P\}$, we write $\Phi \models_P A$. We have strong soundness: for any set $\Phi \cup \{A\} \subseteq \mathsf{Fm}$,

$$\Phi \vdash_L A \Longrightarrow \Phi \models_K A.$$

Figure 3 (Łukasiewicz row). $x*y = \max(x+y-1, 0)$ and $x \to y = 1-x+y$ when $x > y$. We have $x \to y = 1$ when $x \leq y$. Its FLew algebra is $[0,1]_{\mathrm{Ł}}$.

Theorem 10 (Completeness of major fuzzy logics; selected pair). Finite strong completeness holds for $(\mathrm{Ł}, [0,1]_{\mathrm{Ł}})$. Note that implication $\to$ is continuous in $[0,1]_{\mathrm{Ł}}$, while it is discontinuous at $(0,0)$ in $[0,1]_G$ and $[0,1]_\Pi$. In fact, we have $0 \to 0 = 1$, while $x \to 0 = 0$ for any $x > 0$. Since any continuous t-norm is an ordinal sum of Gödel, Łukasiewicz and product norms [26] (see also [15,8,24]), a well-known fact is obtained.

Theorem 12. Among the infinitely many t-norms, Łukasiewicz norm is the only one that makes all logical connectives $\wedge, \vee, \cdot, \to$ continuous.

Author source: homomorphism calculation used again in Theorem 31 (30:10)

Define a (partial) valuation $v : \overline{\Phi} \to [0,1]_{\mathrm{Ł}}$ by setting $v(A_i) := c_i$. It is homomorphic for each $\star \in \{\wedge, \vee, \cdot, \to\}$. In fact, if $A_i \equiv A_j \star A_k$,

$$v(A_i) = c_i = f_i(\mathbf{c}) = c_j \star c_k = v(A_j) \star v(A_k).$$

Author source: naive set theory (30:12)

Let $\mathsf{Var}_{\mathrm{set}} := \{x_0, x_1, x_2, \ldots\}$ be a fixed countable set of term variables. The set $\mathsf{Tm}_{\mathrm{set}}$ of terms and the set $\mathsf{Fm}_{\mathrm{set}}$ of formulas are simultaneously defined by the following grammar:

$$\begin{aligned} t, u &::= x \mid \{x \mid A\}, \\ A, B &::= t \in u \mid 1 \mid 0 \mid A \wedge B \mid A \vee B \mid A \cdot B \mid A \to B, \end{aligned}$$

where variable x is bound in formula $\forall x.A$ and in term $\{x \mid A\}$. Existential quantifier is defined by $\exists y.A := \forall x.(\forall y.(A \to x^\bullet)) \to x^\bullet$, where x is fresh and $x^\bullet := \bullet \in x$ for a fixed closed term $\bullet$. In the sequel, we will also use notation $A \otimes B$ instead of $A \cdot B$ for readability. The sequent calculus for $\mathsf{FLew}_{\mathrm{set}}$ consists of the inference rules in Figure 1 together with:

$$\frac{A[t/x], \Delta \Rightarrow \Pi}{t \in \{x \mid A\}, \Delta \Rightarrow \Pi}\ (\in \ell) \qquad \frac{\Gamma \Rightarrow A[t/x]}{\Gamma \Rightarrow t \in \{x \mid A\}}\ (\in r)$$

$$\frac{A[t/x], \Delta \Rightarrow \Pi}{\forall x.A, \Delta \Rightarrow \Pi}\ (\forall \ell) \qquad \frac{\Gamma \Rightarrow A}{\Gamma \Rightarrow \forall x.A}\ (\forall r)$$

where rule $(\forall r)$ is subject to the eigenvariable condition. Given a set $\Phi \cup \{A\} \subseteq \mathsf{Fm}_{\mathrm{set}}$, we write $\Phi \vdash_{\mathsf{FLew}_{\mathrm{set}}} A$ if A^{seq} is derivable from Φ^{seq} by the above inference rules. Given $L \in \mathrm{Ext}(\mathsf{FLew})$, let $\widetilde{L} \subseteq \mathsf{Fm}_{\mathrm{set}}$ be the set of all substitution instances of propositional formulas in L (in the language $\mathsf{Fm}_{\mathrm{set}}$). We define $L_{\mathrm{set}} := \{A \in \mathsf{Fm}_{\mathrm{set}} \mid \widetilde{L} \vdash_{\mathsf{FLew}_{\mathrm{set}}} A\}$, and write $\vdash_{L_{\mathrm{set}}} A$ if $A \in L_{\mathrm{set}}$.

Author source: restricted system (30:15)

What we will now take is the second approach. To apply the BFT argument, we have to find a suitable fragment of Ł_{set}, for which a hypercube of finite dimension suffices. The following fragment meets this requirement.

Definition 30. System $\mathsf{FLew}^{-}_{\text{set}}$ is obtained from $\mathsf{FLew}_{\text{set}}$ by imposing the following restriction:

> (*) when applying rule $(\forall\ell)$, t must be either a variable or a closed term.

We define $\text{Ł}^{-}_{\text{set}} := \{A \in \mathsf{Fm}_{\text{set}} \mid \widetilde{\text{Ł}} \vdash_{\mathsf{FLew}^{-}_{\text{set}}} A\}$.

Author source: complete consistency construction and proof (30:15–16)

Let us now prove the consistency of $\text{Ł}^{-}_{\text{set}}$. We begin with some preparations. Suppose that a finite set $\Phi \subseteq \mathsf{Fm}_{\text{set}} \cup \mathsf{Tm}_{\text{set}}$ of formulas and terms is given. We consider its closure $\overline{\Phi}$ under subformulas and subterms. Explicitly, $\overline{\Phi}$ is the least set such that (i) $\Phi \subseteq \overline{\Phi}$, (ii) $A \star B \in \overline{\Phi}$ implies $A, B \in \overline{\Phi}$ for $\star \in \{\wedge, \vee, \cdot, \to\}$, (iii) $\forall x.A \in \overline{\Phi}$ implies $A \in \overline{\Phi}$, (iv) $(t \in u) \in \overline{\Phi}$ implies $t, u \in \overline{\Phi}$, and (v) $\{x \mid A\} \in \overline{\Phi}$ implies $A \in \overline{\Phi}$. Let Φ_{ctm} be the set of closed terms in $\overline{\Phi}$, and let

$$\Phi_{\text{cfm}} := \{A\sigma \in \mathsf{Fm}_{\text{set}} \mid A \in \overline{\Phi},\ \sigma : \mathsf{Var}_{\text{set}} \to \Phi_{\text{ctm}}\},$$

where $A\sigma$ is a closed formula obtained by applying substitution σ to A. As is easily observed, $\forall x.B \in \Phi_{\text{cfm}}$ implies $B[t/x] \in \Phi_{\text{cfm}}$ for every $t \in \Phi_{\text{ctm}}$. Since $\overline{\Phi}$ is a finite set, so are Φ_{ctm} and Φ_{cfm}. Suppose that $\Phi_{\text{cfm}} = \{A_1, \dots, A_n\}$. We define a function $f = (f_1, \dots, f_n) : [0,1]^n \to [0,1]^n$ as in the proof of Theorem 14. Concretely, given $\mathbf{x} = (x_1, \dots, x_n) \in [0,1]^n$, let

$$\begin{aligned}
f_i(\mathbf{x}) &:= \ell && \text{if } A_i \equiv \ell \ (\ell = 0, 1),\\
&:= x_j \star x_k && \text{if } A_i \equiv A_j \star A_k \ (\star \in \{\wedge, \vee, \cdot, \to\}),\\
&:= \bigwedge_{t \in \Phi_{\text{ctm}}} x_{j_t} && \text{if } A_i \equiv \forall x.B \text{ and } B[t/x] \equiv A_{j_t} \text{ for each } t \in \Phi_{\text{ctm}} \ (1 \le j_t \le n),\\
&:= x_j && \text{if } A_i \equiv t \in \{x \mid B\} \text{ and } B[t/x] \equiv A_j \text{ for some } 1 \le j \le n,\\
&:= 0 && \text{if } A_i \equiv t \in u \text{ but the previous case does not apply.}
\end{aligned}$$

Then f is continuous by Theorem 12 and finiteness of Φ_{ctm}. By BFT, there is $\mathbf{c} = (c_1, \dots, c_n) \in [0,1]^n$ such that $f(\mathbf{c}) = \mathbf{c}$. This gives us a function $v : \Phi_{\text{cfm}} \to [0,1]_{\text{Ł}}$ defined by $v(A_i) := c_i$. v is homomorphic with respect to propositional connectives $1, 0, \wedge, \vee, \cdot, \to$ as in the proof of Theorem 14. Moreover, if $A_i \equiv t \in \{x \mid B\}$ and $A_j \equiv B[t/x]$, we have

$$v(t \in \{x \mid B\}) = c_i = f_i(\mathbf{c}) = c_j = v(B[t/x]), \tag{5}$$

so v validates rules $(\in \ell)$ and $(\in r)$ applied to formulas in Φ_{cfm}. Finally if $A_i \equiv \forall x.B$ and $A_{j_t} \equiv B[t/x]$ for each $t \in \Phi_{\text{ctm}}$, we have

$$v(\forall x.B) = c_i = f_i(\mathbf{c}) = \bigwedge_{t \in \Phi_{\text{ctm}}} c_{j_t} = \bigwedge_{t \in \Phi_{\text{ctm}}} v(B[t/x]). \tag{6}$$

We are now ready to give a consistency proof.

Theorem 31. $\text{Ł}^{-}_{\text{set}}$ is consistent.

Proof. Suppose that $\text{Ł}^{-}_{\text{set}}$ is inconsistent. This means that there are a finite set of formulas $\Psi \subseteq \widetilde{\text{Ł}}$ and a derivation π of $\Rightarrow 0$ from Ψ^{seq}. Let Φ be the finite set of all formulas occurring in π, and define $\Phi_{\text{ctm}}, \Phi_{\text{cfm}}$ and valuation $v : \Phi_{\text{cfm}} \to [0,1]_{\text{Ł}}$ as above. We prove by induction on the derivation the following:

> if sequent $\Gamma \Rightarrow \Pi$ occurs in π and $\sigma : \mathsf{Var}_{\text{set}} \to \Phi_{\text{ctm}}$, then $v(\Gamma\sigma) \leq v(\Pi\sigma)$.

Here $v(\Gamma) := v(A_1) \cdots v(A_m)$ if $\Gamma \equiv A_1, \ldots, A_m$, and $v(\Gamma) := 1$ if Γ is empty. Likewise, $v(\Pi) := v(A)$ if $\Pi \equiv \{A\}$, and $v(\Pi) := 0$ otherwise.

- If the derivation just consists of $\Rightarrow B$ with $B \in \Psi$, then we have $1 \leq v(B\sigma)$ as $B\sigma$ is a substitution instance of a propositional formula in Ł.
- If the derivation ends with a rule for a propositional connective or $(\in \ell)$ or $(\in r)$, the claim easily follows from the induction hypothesis and (5).
- Suppose that the derivation ends with rule $(\forall r)$ inferring $\Gamma \Rightarrow \forall x.B$ from $\Gamma \Rightarrow B$. Given $\sigma : \mathsf{Var}_{\text{set}} \to \Phi_{\text{ctm}}$, let σ' be a substitution that agrees with σ on $\mathsf{Var}_{\text{set}} \setminus \{x\}$ but leaves x unchanged. By the induction hypothesis and the eigenvariable condition, we have $v(\Gamma\sigma) = v(\Gamma\sigma') \leq v(B\sigma'[t/x])$ for every $t \in \Phi_{\text{ctm}}$. Hence by (6),

$$v(\Gamma\sigma) \leq \bigwedge_{t \in \Phi_{\text{ctm}}} v(B\sigma'[t/x]) = v(\forall x.B\sigma') = v((\forall x.B)\sigma).$$

- Now suppose that the derivation ends with rule $(\forall \ell)$ inferring $\forall x.B, \Gamma \Rightarrow \Pi$ from $B[t/x], \Gamma \Rightarrow \Pi$. By the induction hypothesis we have $v(B[t/x]\sigma) \cdot v(\Gamma\sigma) \leq v(\Pi\sigma)$. If t is closed, then $t\sigma = t \in \Phi_{\text{ctm}}$. If t is a variable y, then $t\sigma = \sigma(y) \in \Phi_{\text{ctm}}$. In any case, we have $t\sigma \in \Phi_{\text{ctm}}$. So

$$v((\forall x.B)\sigma) = v(\forall x.B\sigma') = \bigwedge_{u \in \Phi_{\text{ctm}}} v(B\sigma'[u/x]) \leq v(B\sigma'[t\sigma/x]) = v(B[t/x]\sigma),$$

  where σ' is as above. Hence we conclude $v((\forall x.B)\sigma) \cdot v(\Gamma\sigma) \leq v(\Pi\sigma)$.

This is a contradiction because derivation π ends with sequent $\Rightarrow 0$. ◁

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